

DATABASE SYSTEMS-BIT1223/BTL1223/BCL1223

TUTORIAL 1

Question 1:

- a) Define data and information. (2 marks)
- b) Explain the relationship between data and information using a business example. (4 marks)
- c) Why is accurate and timely information important for organizational decision-making? (4 marks)

Question 2:

- a) Define a database and a Database Management System (DBMS). (4 marks)
- b) Explain the role of a DBMS as an intermediary between users and databases. (4 marks)
- c) State and explain any **THREE(3)** advantages of using a DBMS. (6 marks)

Question 3:

- a) Differentiate between a single-user database and a multiuser database. (4 marks)
- b) Compare a centralized database and a distributed database. (4 marks)
- c) A multinational company operates branches in several countries. Recommend the most suitable database type and justify your answer. (7 marks)

Question 4:

- a) Describe two characteristics of traditional file systems. (4 marks)
- b) Explain three limitations of file systems that led to the development of DBMSs. (6 marks)
- c) Discuss how a DBMS overcomes these limitations. (5 marks)

Question 5:

- a) Define data redundancy and data inconsistency. (4 marks)
- b) Explain how data redundancy can lead to data inconsistency. (4 marks)
- c) Provide a real-world example illustrating this problem. (4 marks)

Question 6:

- a) What are data anomalies? (2 marks)
- b) Explain Update, Insertion, and Deletion anomalies with suitable examples. (9 marks)

c) Suggest how database design can minimize these anomalies. (4 marks)

Question 7:

a) Describe the Conceptual, Logical, and Physical Data Models. (9 marks)

b) Explain why data modeling skills are important during database development. (4 marks)

Question 8:

a) Identify the five major components of a database system. (5 marks)

b) Explain the role of each component in a university database system. (10 marks)

Question 9:

a) Explain the purpose of a data dictionary in a DBMS. (4 marks)

b) Describe any four DBMS functions. (8 marks)

c) Why are backup and recovery functions critical in modern organizations? (3 marks)

Question 10:

a) Identify and explain **THREE(3)** problems caused by a file-based system. (6 marks)

b) Recommend a database solution. (2 marks)

c) Explain how a DBMS improves data sharing, integrity, and security. (9 marks)

d) Discuss **ONE(1)** disadvantage of implementing a database system and how it can be addressed. (3 marks)